

File Edit Format Options Window Execute ?

exp 6.sci (C:\Users\pach\Documents\exp 6.sci) - SciNotes

exp 6.sci exp 7.sci exp 8.sci exp 9.sci

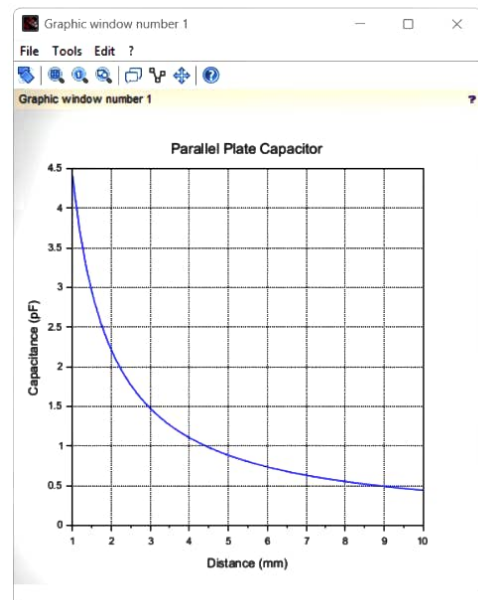
```
1 clc;
2 clear;
3
4 // Constant
5 e = 8.854e-12;
6
7 //-----
8 // Parallel Plate Capacitor
9 //-----
10
11 // Given values
12 a1 = 5e-4; // 5 cm^2 -> m^2
13 d = 3e-3; // 3 mm -> m
14
15 // Capacitance
16 c = e * a1 / d;
17
18 disp("Parallel Plate Capacitor");
19 disp(c, "Capacitance (F) =");
20 disp(c*1e12, "Capacitance (pF) =");
21
22 // Plot
23 x = linspace(0.001,0.01,50); // Distance (m)
24 y = e*a1./x;
25
26 figure(1);
27 plot(x*1000,y*1e12);
28 xlabel("Distance (mm)");
29 ylabel("Capacitance (pF)");
30 title("Parallel Plate Capacitor");
31 xgrid();
32
33 //-----
34 // Multi-Dielectric Capacitor
35 //-----
36
37 a2 = 11e-4; // 11 cm^2 -> m^2
38
39 a1 = 5e-4; // 5 cm^2 -> m^2
```

SciLab 2025.1.0 Console

File Edit Control Applications ?

SciLab 2025.1.0 Console

```
"Parallel Plate Capacitor"
1.476D-12
"Capacitance (F) ="
1.4756667
"Capacitance (pF) ="
" "
"Multi-Dielectric Capacitor"
6.151D-12
"Capacitance (F) ="
6.1512
"Capacitance (pF) ="
--> |
```



exp 6:

~~$a_1 = 3 \text{ cm}^2$~~
 ~~$a_2 = 4 \text{ cm}^2$~~
 ~~$d = 3$~~

$a_1 = 5 \text{ cm}^2$

$a_2 = 11 \text{ cm}^2$

$d = 3 \text{ mm}$

$d_2 = 2 \text{ mm}$

$d_3 = 1 \text{ mm}$

$\epsilon_{r1} = 4$

$\epsilon_{r2} = 3$

$\epsilon_{r3} = 6$

Parallel plate capacitor

$$C = \frac{\epsilon_0 A}{d}$$

$$C = \frac{8.854 \times 10^{-12} \times 5 \times 10^{-4}}{3 \times 10^{-3}}$$

$$C = 1.4757 \times 10^{-12} \text{ F}$$

Multi Dielectric capacitor

$$C = \frac{\epsilon_0 A}{\frac{d_1}{\epsilon_{r1}} + \frac{d_2}{\epsilon_{r2}} + \frac{d_3}{\epsilon_{r3}}}$$

o/p 2V

Δm

$$= \frac{0.003}{4} + \frac{0.002}{3} + \frac{0.001}{6}$$

$$= 0.00075 + 0.000667 + 0.0001667$$

$$= 0.0015834$$

$$C = \frac{9.7394 \times 10^{-16}}{0.0015834}$$

$$C = 6.15 \times 10^{-12} \text{ F}$$